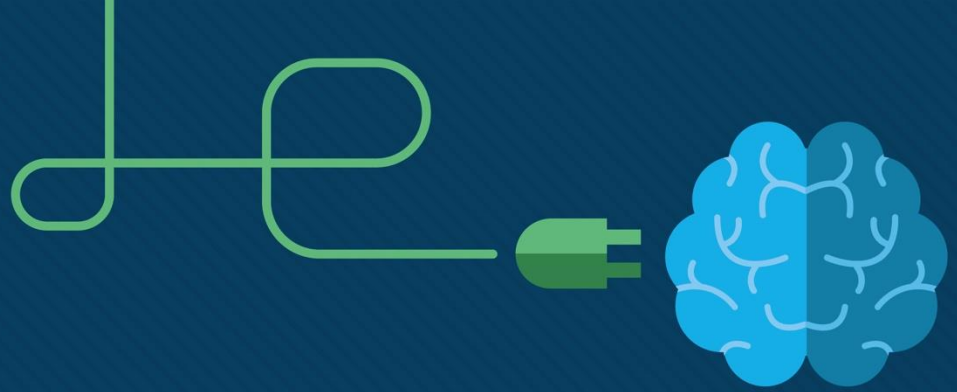

Module 1
Advanced Routing and Switching

ITNET03

Advanced Routing and Switching

Module Objectives

Module Title: Advanced Routing and Switching

Module Objectives:

- Be familiar with the outcomes and policies of the ITNET03 course
- Explain the factors that drive the need to scale a network and the effects to a hierarchical network topology
- Be familiar with the basic technologies used to improve network performance, fault-tolerance, accessibility and manageability

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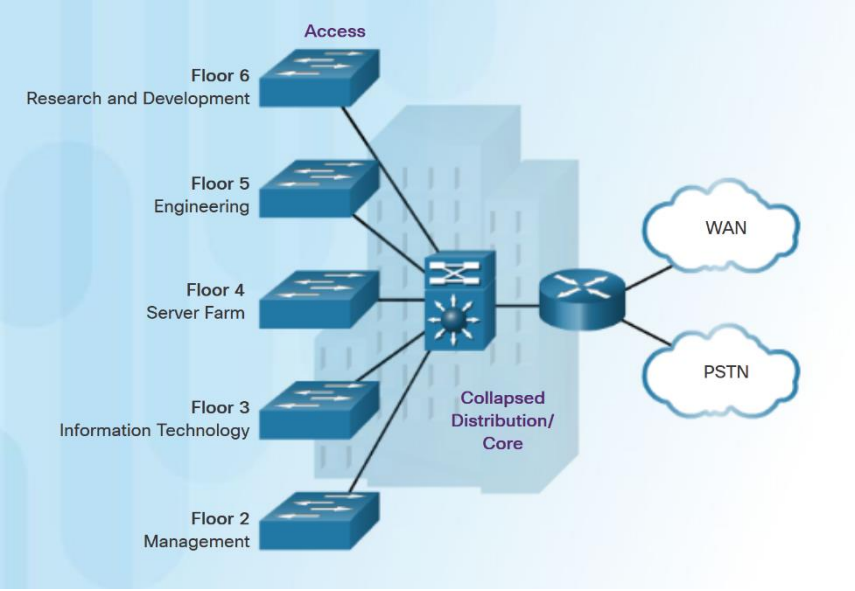
Growing a Network

The Need to Scale the Network



- All enterprise networks must:
  - Support the exchange of various types of network traffic
  - Support critical applications
  - Support converged network traffic
  - Support diverse business needs
  - Provide centralized administrative control
- The LAN is the networking infrastructure that provides access to network resources for end users over a single floor or a building.

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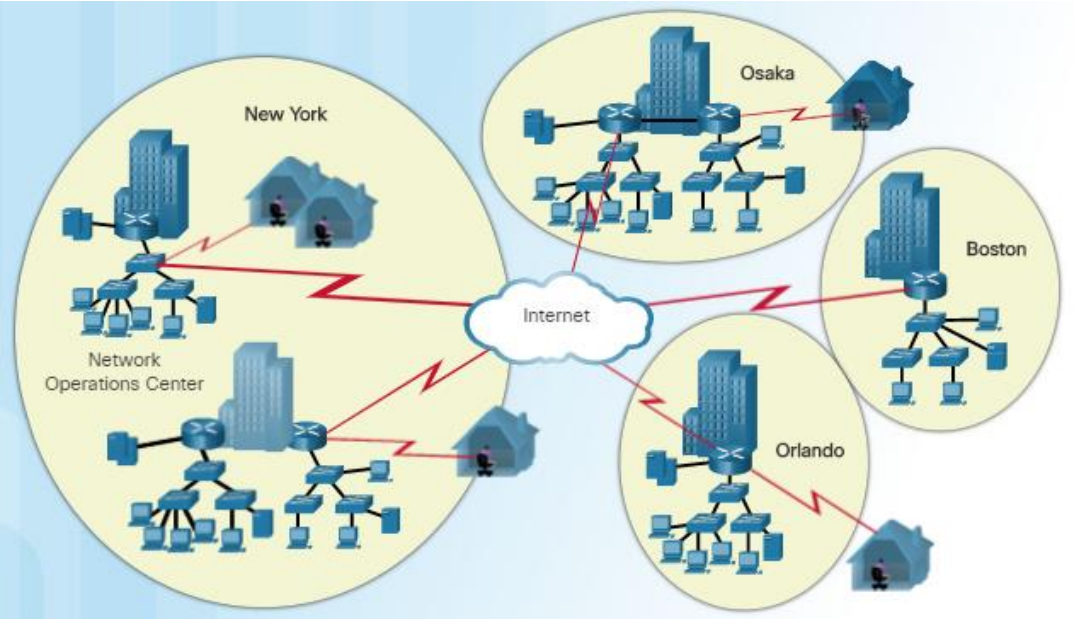
The Need to Scale the Network



- A company with a small network with one site and a connection to the Internet might grow into an enterprise with a central location with numerous remote sites across the globe.

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This Course

ITNET01	ITNET02
Network Protocols Basic LAN device configuration How networks work How to set up a **functional** network	Intermediate switch and router configuration for LAN connectivity How to lay down the foundations of a **properly designed** LAN for a SOHO

ITNET03	ITNET04
Intermediate switch and router configuration for LAN connectivity How to **properly design a network for expansion** to a robust campus LAN	Intermediate switch and router configuration for WAN connectivity Multi-site network

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Learning Outcomes

LO1: Apply knowledge of organizational needs and policies, as well network routing, management and fault-tolerance mechanisms in consideration of their advantages and limitations to conceptualize a network model from defined requirements IT 1.2, 1.3

LO2: Analyze a given scenario and determine the necessary network components to address infrastructure needs using domain knowledge on advanced routing and switching techniques IT 4.2

LO3: Design and implement a single-site campus network exhibiting the appropriate characteristics and functionality for a given scenario IT 6.2, 7.2

LO4: Contribute positively to a team and monitor its progress in achieving a common goal IT 8.3, 11.2

LO5: Effectively document and present completed work IT 9.2, 9.3

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Design for Scalability

- The network designer must develop a strategy to enable the network to available, scalable and manageable
  - Use expandable, modular equipment that can be easily upgraded to increase capabilities and capacity.
  - Design a hierarchical network to include modules that can be added, upgraded, and modified as needed.
  - Create an IPv4 or IPv6 address strategy that is hierarchical.
  - Choose routers or multilayer switches to limit broadcasts and filter undesirable traffic from the network.
  - Implement redundant links between critical devices and between access and core layers.
  - Incorporate wireless LAN devices to support user mobility
  - Use network management tools to enable centralized monitoring and administration

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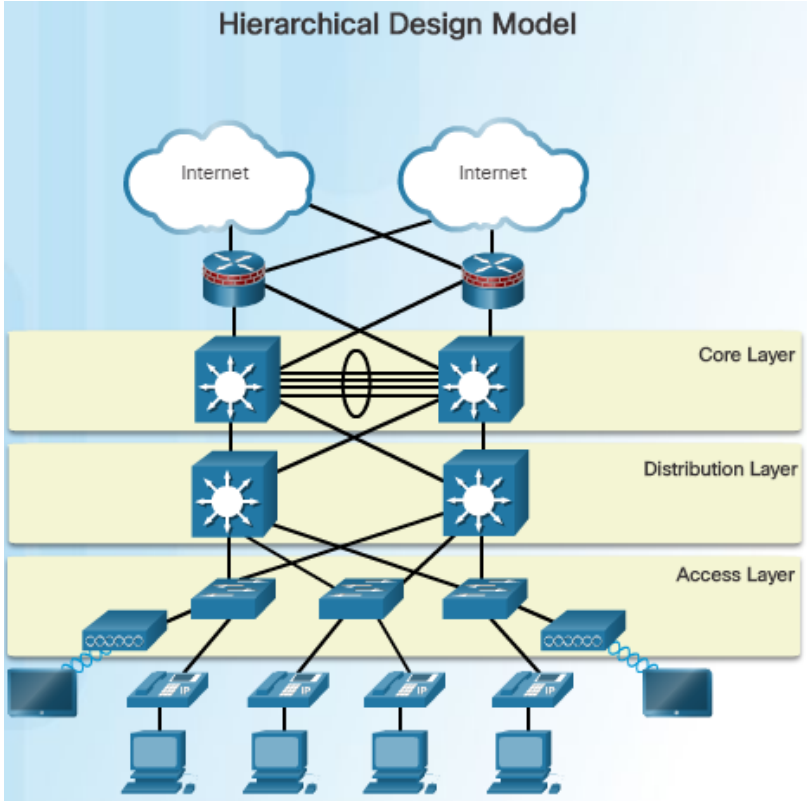
Maintaining Hierarchy



- The campus wired LAN uses a hierarchical design model to break the design up into modular layers.
- Breaking the design up into layers allows each layer to perform specific functions, which simplifies the network design for easier deployment and management.
- A hierarchical LAN design includes three layers as shown in the figure:
  - Access layer
  - Distribution layer
  - Core layer
- Some smaller enterprise networks implement a two-tier hierarchical design and collapse the core and distribution layers into one layer.

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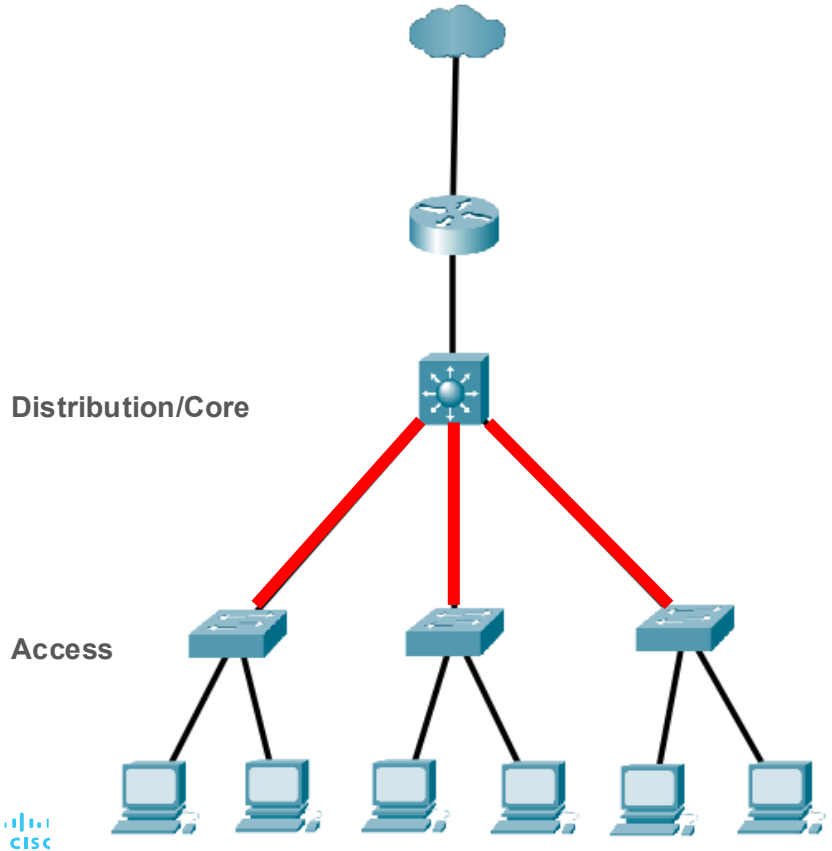
Increasing Bandwidth



- In a hierarchical network design, some links between access and distribution layer switches may need to process a greater amount of traffic than other links do.
- As multiple links converge into a single link, it is possible for this link to become a bottleneck.

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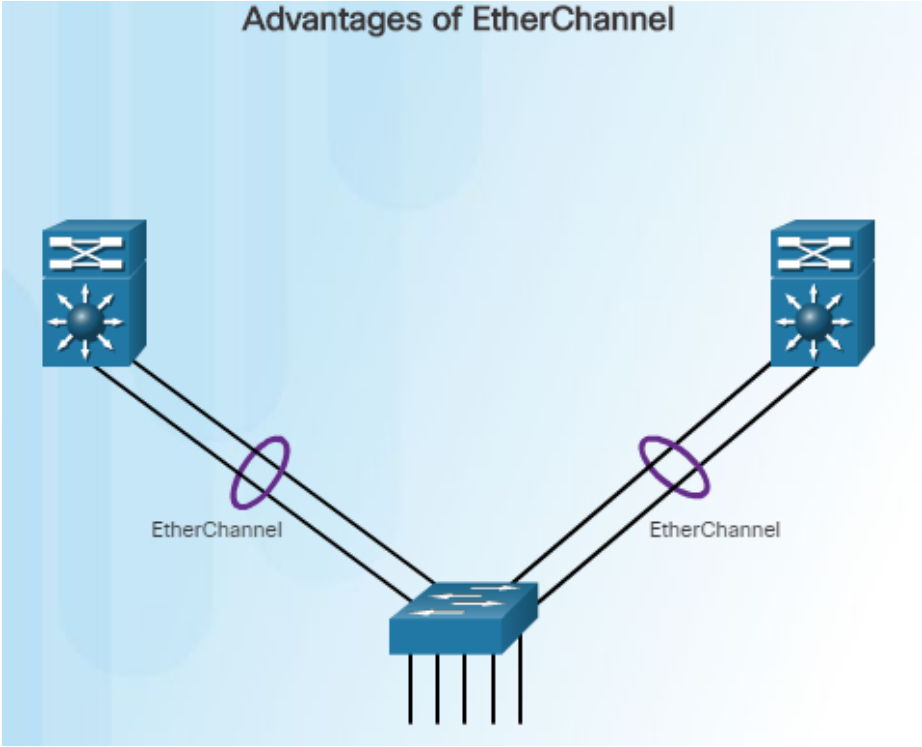
Increasing Bandwidth



- Link aggregation allows the network administrator to increase the amount of bandwidth between devices
  - Several physical links can be bundled together into a single high bandwidth logical link.
  - The logical link can take advantage of load balancing between the physical links
- Cisco's link aggregation technology is branded as 'EtherChannel'

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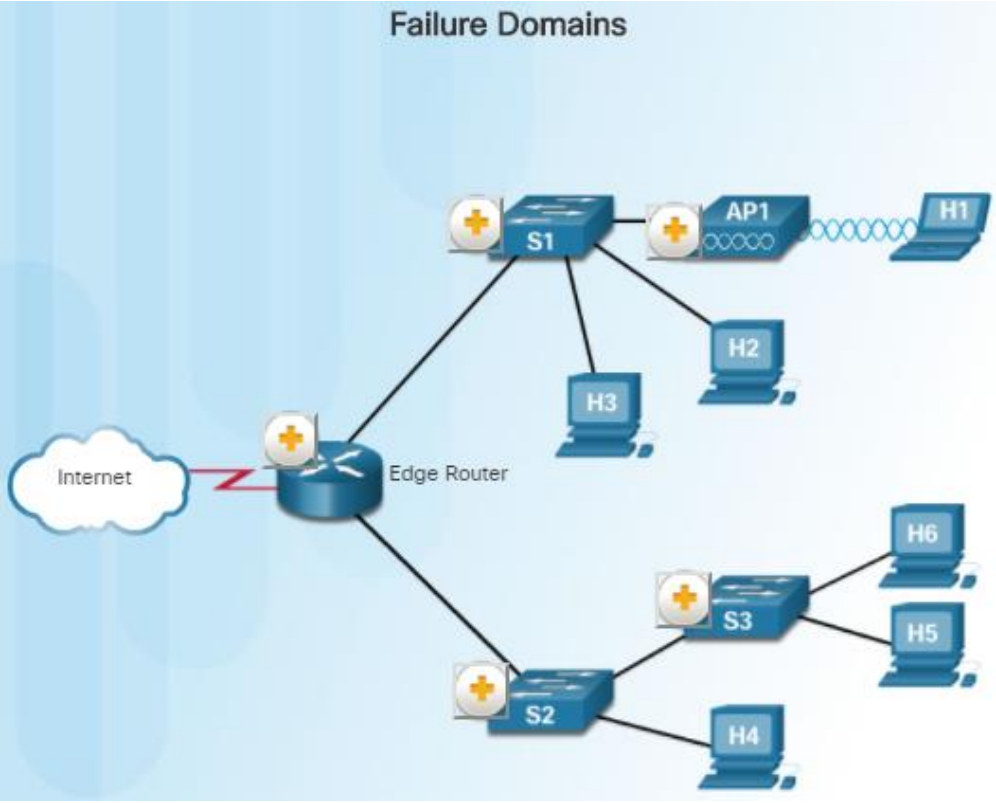
Failure Domains



- A well-designed network should limit the size of failure domains.
- A failure domain is the area of a network that is impacted when a critical device or network service experiences problems.
- The function of the devices that fail will determine the impact of the failure domain.
- Smaller failure domains reduce the impact of a failure but also make troubleshooting easier.

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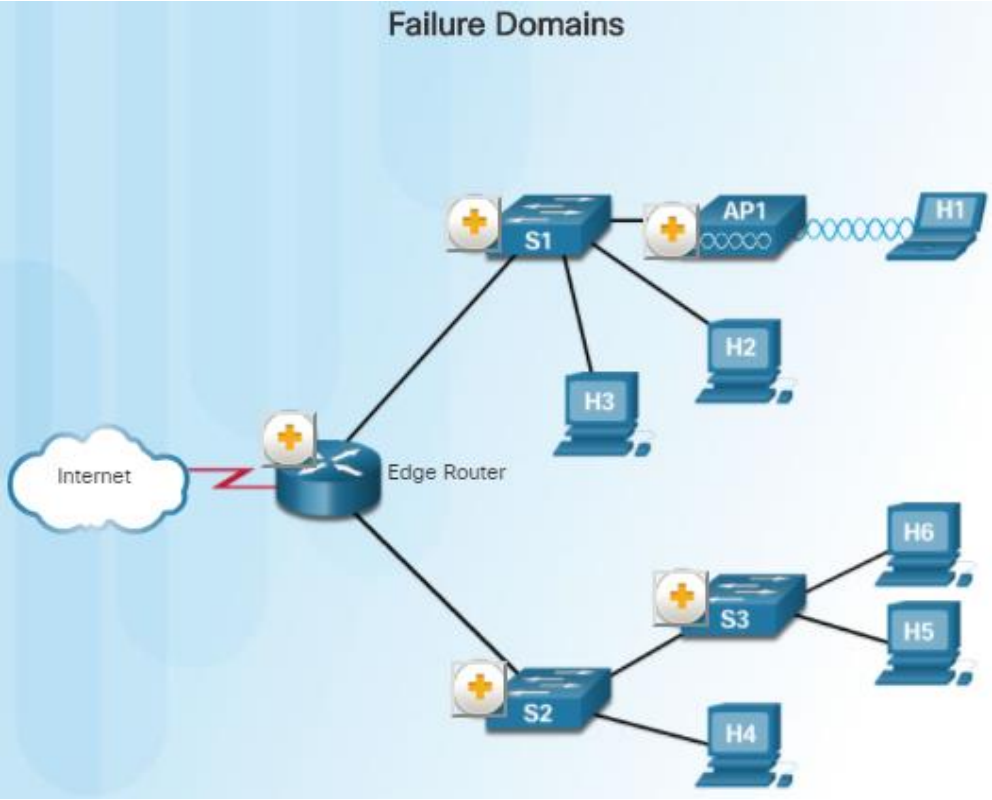
Failure Domains (Cont.)



- In the hierarchical design model, it is usually easier to control the size of a failure domain in the distribution layer.
- In the distribution layer, network errors can be contained to a smaller area which will impact fewer users.
- When using Layer 3 devices at the distribution layer, every router functions as a gateway for a limited number of access layer users.
- Switch Block Deployment
  - Routers or multilayer switches are usually deployed in pairs with access layer switches evenly divided between them.
  - Each switch block acts independently of the others, which reduces the impact of failures.

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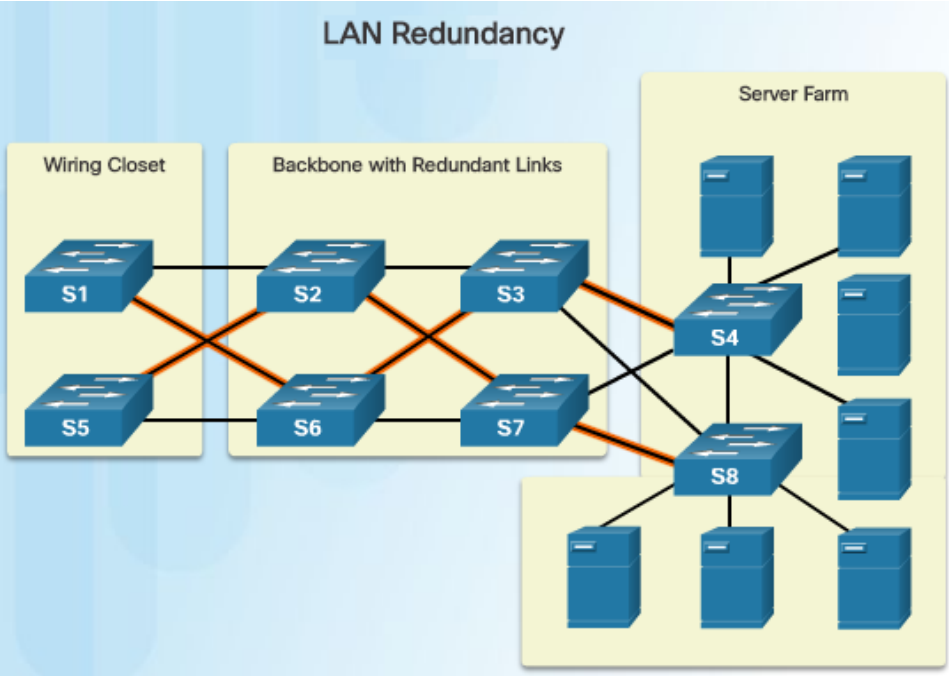
Planning for Redundancy



- Redundancy is an important part of the network design for preventing disruption of network services.
- Minimize the possibility of a single point of failure by recognizing these facts:
  - Installing duplicate equipment and providing failover services for critical devices is necessary.
  - Installing redundant paths to offer alternate physical paths for data to traverse the network.

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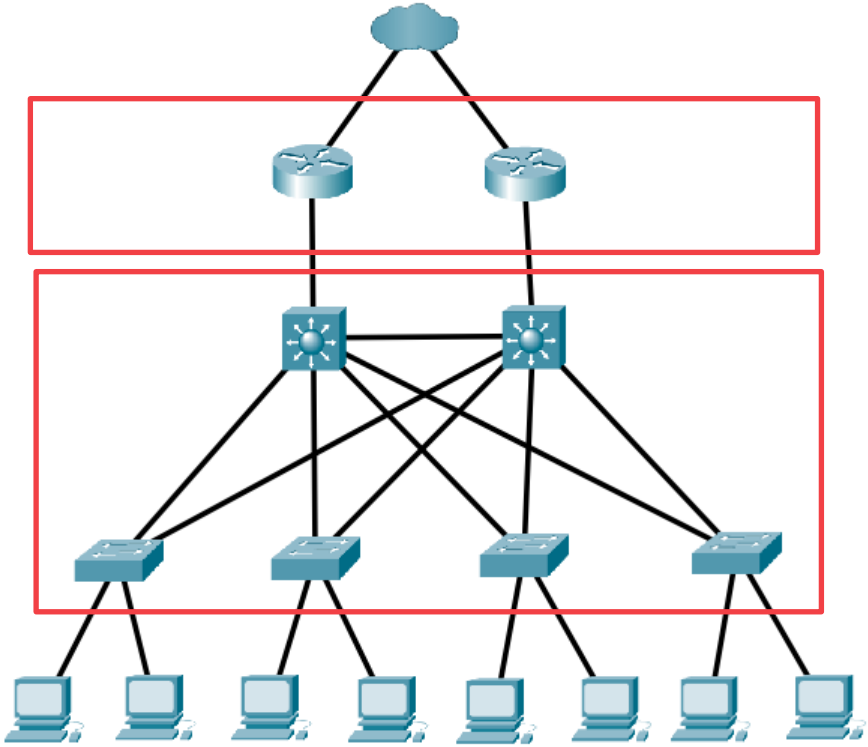
Implementing Redundancy



- Spanning Tree Protocol (STP) provides a mechanism for disabling redundant paths in a switched network until the path is necessary such as when a failure occurs.
- First Hop Redundancy Protocols (FHRP) enable coordination between duplicate routers to adjust to a gateway failure

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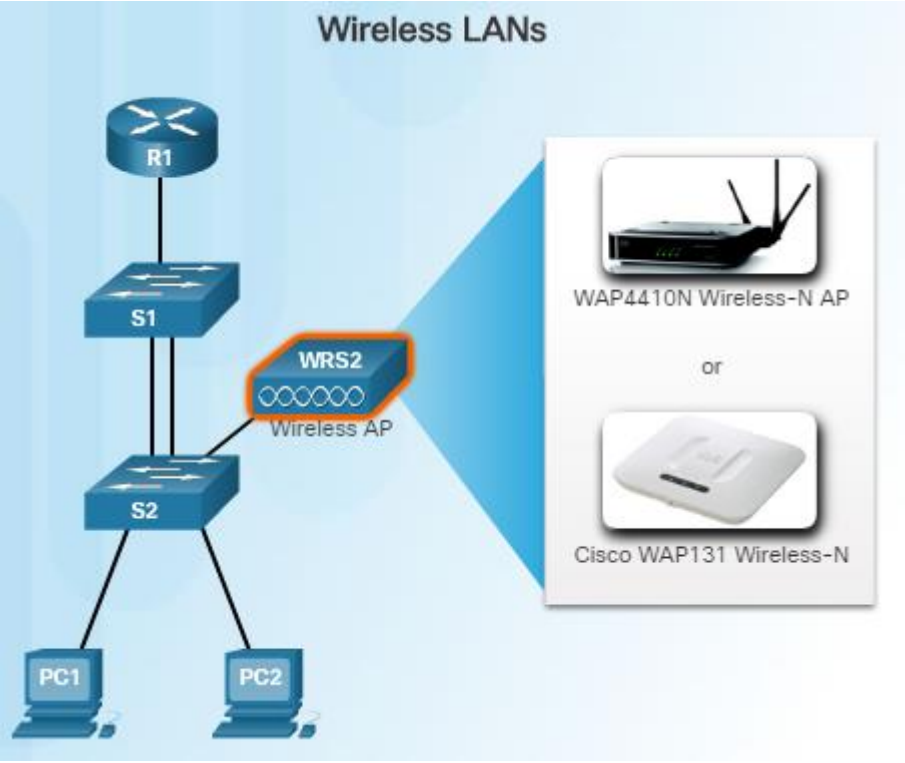
Expanding the Access Layer



- The network must be designed to be able to expand network access to individuals and devices, as needed.
- Wireless connectivity is an important aspect of extending access layer connectivity by supporting user mobility.
- Advantages of wireless connectivity include increased flexibility, reduced cost, and the ability to adapt to changing network and business requirements.
- End devices can use a wireless NIC that incorporates a radio transmitter/receiver and appropriate software drivers to connect to a WLAN

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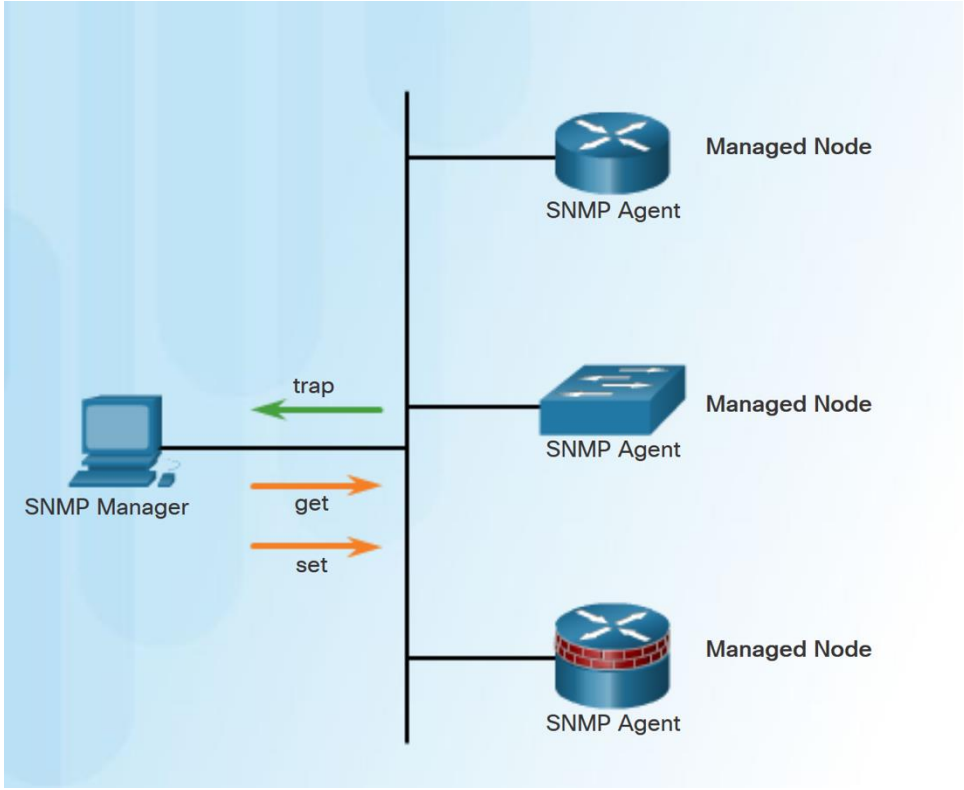
Network Management



- As the network grows, an administrator needs to manage an increasing number and more varied types of infrastructure devices
- Network management involves
  - Collecting and analyzing data and events in day-to-day network operations
  - Implementing configuration changes to network devices
- Network management tools simplify administrative tasks by providing centralized and automated management capabilities to network administrators for a wide variety of devices

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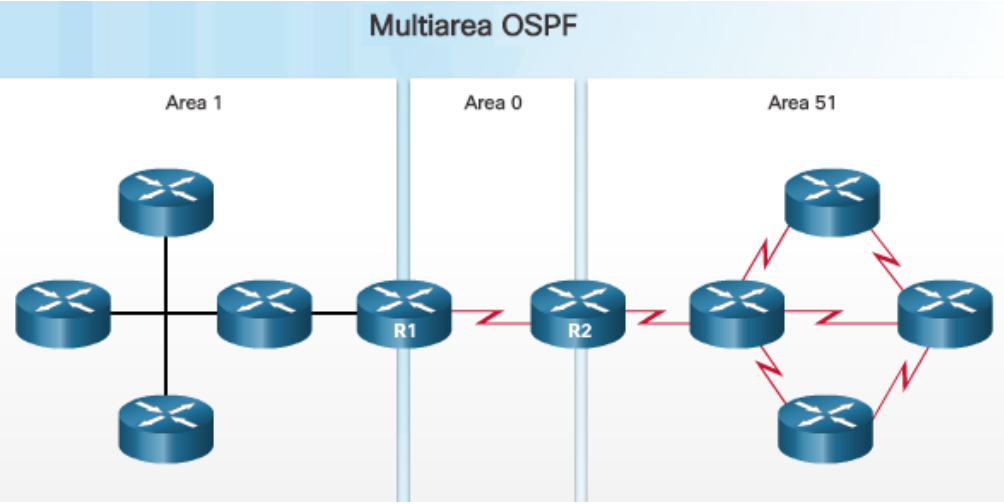
Packet Routing



- As a LAN grows, routers may be used to connect one network to another network and determine the best route for traffic to the destination
- Routers may learn routes
  - statically through manual administrator configuration
  - or dynamically by automatic sharing of routes with other routers using a routing protocol
- An essential skill for scaling networks is formulating and interpreting network routes to move packets,

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What did you learn in this module?

- Scaling a network correctly relies on a solid, hierarchical design foundation
- When growing a LAN, performance, scalability, availability and manageability become critical factors to consider
- For this course, the following topics will be covered to address the requirements of a growing network:
  - Performance: Link aggregation
  - Availability: Redundancy through spanning tree and first hop redundancy
  - Scalability: Wireless LANs, routing
  - Manageability: Network management protocols

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About This Course

Course Outline

Week	Topics and Readings
1	How should a LAN be designed to easily scale and minimize impact of component failure? Advanced Routing and Switching 1. Growing a Network
1-2	How can a network be designed to recover automatically from link failure? Spanning Trees 2.1 Spanning Tree Concepts 2.2 Spanning Tree Protocol Varieties 2.3 Spanning Tree Configuration
3	How can multiple switch links be simultaneously used to aggregate available network bandwidth Link Aggregation 3.1 Link Aggregation Concepts 3.2 Link Aggregation Configuration
3-4	How can a network be designed to recover automatically from gateway failure? Gateway Redundancy 4.1 First Hop Redundancy Principles 4.2 Hot Standby Router Protocol Configuration
5	How do wireless LANs operate and promote network mobility? Wireless Networks 5.1 Wireless Concepts and Operation 5.2 Wireless Lan Security 5.3 Wireless LAN Configuration
6	What are the different tools available to simplify network monitoring? Network Management 6.1 Device Discovery 6.2 Device Management 6.3 Device Maintenance
7-8	How does a router make data forwarding decisions? Routing Concepts 7.1 Path Determination 7.2 The Routing Table 7.3 Static vs Dynamic Routing
10-11	How are static routes used and how are they configured? Static Routing 8.1 Introduction to Static Routing 8.2 Configure Static Routes 8.3Route Summarization 8.4 Static Route Troubleshooting
11-12	How is a dynamic routing protocol used to build routing tables? Open Shortest Path First 8.1 Link State Dynamic Routing 8.2 OSPF Operation 8.3 Single Area OSPF Configuration 8.4 OSPF Fine Tuning

Class Requirements and Learning Methods:

ITNET03 and LBYITN3

Requirement/Assessment Task	Weight
Departmental Exams Theory Skills	 20% 25%
Assignments and Seatwork	10%
Laboratory Experiments	25%
Network Design Project	20%
TOTAL	100%

Passing grade = 70%
*Same final grade for lecture and lab

Synchronous
Online Lecture
Consultation
Reading Assignments
Simulated Lab Activities
Group Activities

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Some House Rules

- On online conferencing
  - Synchronous sessions will start promptly
  - Please keep microphones and webcams off unless the activity calls for their use
  - Do participate in interactive parts of the sessions and feel free to ask questions (you may use your microphone or type it in the chat)

- On class activities
  - Start working on them as soon as possible when they're given
  - Deadlines will usually be set on the next class meeting (more days for longer assignments)
  - Can be made up within 3 days only after the deadline if missed (but subject to deductions)
    - -10% per day if submitted after the deadline (e.g. deadline Sept 2, submitted Sept 4 = -20%)
  - Late work will NOT BE ACCEPTED more than 3 days after the deadline

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Some House Rules

- **Academic Honesty**
  - Rely on your own effort when working on course deliverables
  - You may use the Internet as a reference for learning activities, but **never** directly copy stuff from it,
  - Any answers sourced from references must be properly cited
  - Use of generative AI tools is prohibited in this class for any graded work unless otherwise stated
  - Copied work = 0 credit (with possible failure for the course and disciplinary action depending on gravity of offense)

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Some House Rules

- **Exams**
  - Exams will be given face-to-face during scheduled dates
  - Independent work. You are expected to practice academic honesty
  - Inform your teacher ASAP if you cannot take the exam on schedule due to valid reasons (e.g. conflicts with other classes, sickness)

- **Lab Use**
  - Be careful with equipment → If you break it; you replace it.
  - Never bring equipment out of the lab without permission
  - CLAYGO → Always clear configurations and place equipment/cables neatly back into their storage spaces

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Some House Rules

- On final grades
  - Non-submission of class requirements and missed deadlines can lead to failure
  - Drop by the 7th week of classes if you think you cannot comply with class requirements.
  - No requests for extra credit work or make up for long overdue class requirements especially at term-end

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